Apache Spark

Spark data sources/sinks

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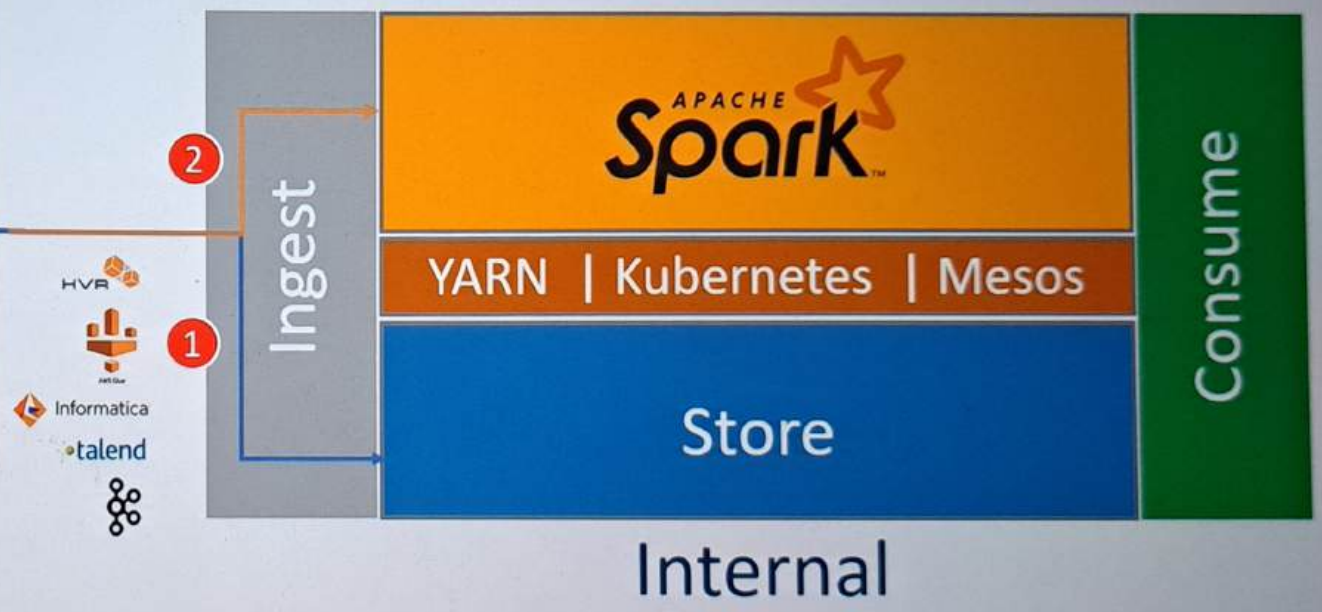
Data Lake

External

1. **JDBC Data Sources**
   Oracle, SQL Server, PostgreSQL
2. **No SQL Data Systems**
   Cassandra, MongoDB
3. **Cloud Data Warehouses**
   Snowflake, Redshift
4. **Stream Integrators**
   Kafka, Kinesis



Internal

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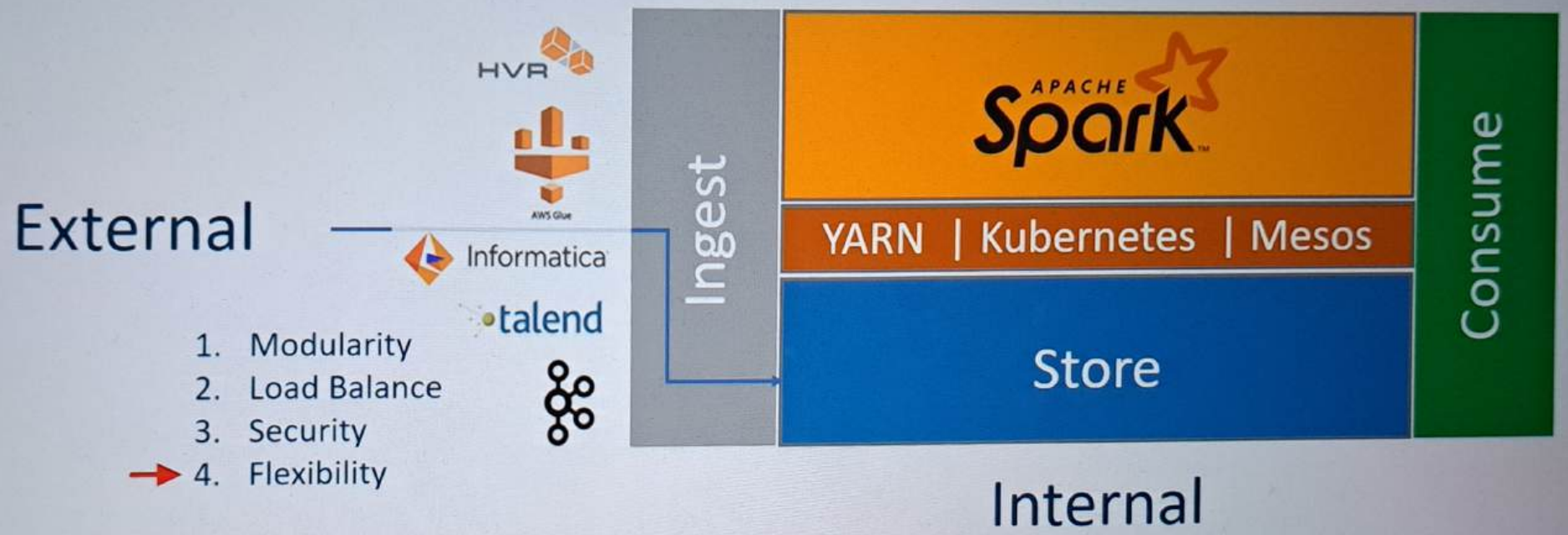
Data Lake



1. Modularity
2. Load Balance
3. Security
4. Flexibility

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Data Lake

External

1. JDBC Data Sources
2. NoSQL Data Systems
3. Cloud Data System
4. Stream Integrators

External

1. JDBC Data Sources
2. NoSQL Data Systems
3. Cloud Data System
4. Stream Integrators

Internal

1. CSV
2. JSON
3. Parquet
4. AVRO
5. Plain Text

1. Spark SQL Tables
2. Delta Lake

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Apache Spark

Spark Data Source API

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Spark Data Source API

DataFrameReader API : https://spark.apache.org/docs/latest/api/python/pyspark.sql.html?highlight=dataframereader#pyspark.sql.DataFrameReader

General structure

```
DataFrameReader
    .format(...)
    .option("key", "value")
    .schema(...)
    .load()
```

Indicative Example

```
spark.read
    .format("csv")
    .option)"header", "true")
    .option("path", "/data/mycsvfiles/")
    .option("mode", "FAILFAST")
    .schema(mySchema)
    .load()
```

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Read Mode
1. PERMISSIVE
2. DROPMALFORMED
3. FAILFAST

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Schema
1. Explicit
2. Infer Schema
3. Implicit

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Apache Spark

Using Spark DataFrameReader

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Spark DataFrameReader

In this video

1. How to use DataFrameReader for CSV, JSON, and Parquet

In next Video

1. Spark Data Types
2. How to explicitly define a schema for your data